Pythagorean Theorem in Real-World Problems

Question Bank 6.04

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 7, sketch the right triangle first. Round to the nearest tenth when the answer is not a whole number.

1. A TV screen is 9 in tall and 12 in wide. How long is its diagonal?

2. A 14-ft ladder leans on a wall with its foot 3 ft from the wall. How high does it reach?

3. A ramp rises 2 m over a run of 9 m. How long is the ramp?

4. A field is 6 m by 8 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides?

5. Will a 12-in ruler lie flat in a box 8 in by 9 in, corner to corner?

6. A kite's 13-m string is held at a spot 5 m from the point straight below the kite. How high is the kite?

7. A wire runs from the top of an 8-m pole to a stake in the ground 6 m from the pole's base. How long is the wire?

Choose one.

8. A 13-ft ladder reaches 12 ft up a wall. How far from the wall is its foot?

- ○ A. 1 ft
- ○ B. 5 ft
- ○ C. 17.7 ft
- ○ D. 25 ft

9. A ramp rises 3 ft over a run of 20 ft. Which length is the hypotenuse?

- ○ A. the run
- ○ B. the rise
- ○ C. the ramp
- ○ D. none of them